

2.3 The Formal Viewpoint

Sometimes it is good to step back from the forest of details and look for general patterns. In this rather brief section we will first describe the general pattern of homology by axioms, then we will look at some common formal features shared by many of the constructions we have made, using the language of categories and functors which has become common in much of modern mathematics.

Axioms for Homology

For simplicity let us restrict attention to CW complexes and focus on reduced homology to avoid mentioning relative homology. A (reduced) **homology theory** assigns to each nonempty CW complex X a sequence of abelian groups $\tilde{h}_n(X)$ and to each map $f: X \rightarrow Y$ between CW complexes a sequence of homomorphisms $f_*: \tilde{h}_n(X) \rightarrow \tilde{h}_n(Y)$ such that $(fg)_* = f_*g_*$ and $\mathbb{1}_* = \mathbb{1}$, and so that the following three axioms are satisfied.

- (1) If $f \simeq g: X \rightarrow Y$, then $f_* = g_*: \tilde{h}_n(X) \rightarrow \tilde{h}_n(Y)$.
- (2) There are boundary homomorphisms $\partial: \tilde{h}_n(X/A) \rightarrow \tilde{h}_{n-1}(A)$ defined for each CW pair (X, A) , fitting into an exact sequence

$$\cdots \xrightarrow{\partial} \tilde{h}_n(A) \xrightarrow{i_*} \tilde{h}_n(X) \xrightarrow{q_*} \tilde{h}_n(X/A) \xrightarrow{\partial} \tilde{h}_{n-1}(A) \xrightarrow{i_*} \cdots$$

where i is the inclusion and q is the quotient map. Furthermore the boundary maps are natural: For $f: (X, A) \rightarrow (Y, B)$ inducing a quotient map $\bar{f}: X/A \rightarrow Y/B$, there are commutative diagrams

$$\begin{array}{ccc} \tilde{h}_n(X/A) & \xrightarrow{\partial} & \tilde{h}_{n-1}(A) \\ \downarrow \bar{f}_* & & \downarrow f_* \\ \tilde{h}_n(Y/B) & \xrightarrow{\partial} & \tilde{h}_{n-1}(B) \end{array}$$

- (3) For a wedge sum $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \hookrightarrow X$, the direct sum map $\bigoplus_{\alpha} i_{\alpha*}: \bigoplus_{\alpha} \tilde{h}_n(X_{\alpha}) \rightarrow \tilde{h}_n(X)$ is an isomorphism for each n .

Negative values for the subscripts n are permitted. Ordinary singular homology is zero in negative dimensions by definition, but interesting homology theories with nontrivial groups in negative dimensions do exist.

The third axiom may seem less substantial than the first two, and indeed for finite wedge sums it can be deduced from the first two axioms, though not in general for infinite wedge sums, as an example in the Exercises shows.

It is also possible, and not much more difficult, to give axioms for unreduced homology theories. One supposes one has relative groups $h_n(X, A)$ defined, specializing to absolute groups by setting $h_n(X) = h_n(X, \emptyset)$. Axiom (1) is replaced by its